



Wheel-Rail Contact: an introduction

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B O S T O N



Overview

- Getting “Inside” the Wheel-Rail Interface
- Fundamental #1: Contact Pressure & Geometry
- Fundamental #2: Traction & Creepage
- Reality Check: The Third Body Layer
- Steering Forces: Putting it all together
- A Special Case: Wheel-Rail Contact and Restraining (Guard) Rails

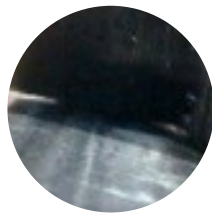
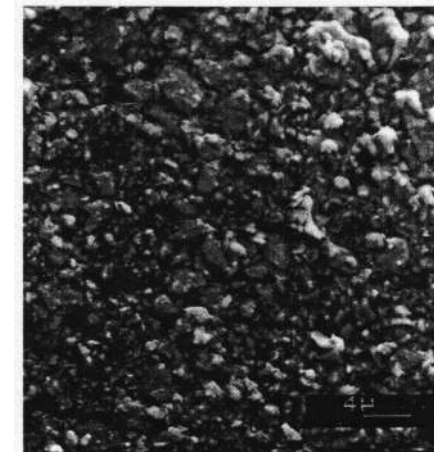
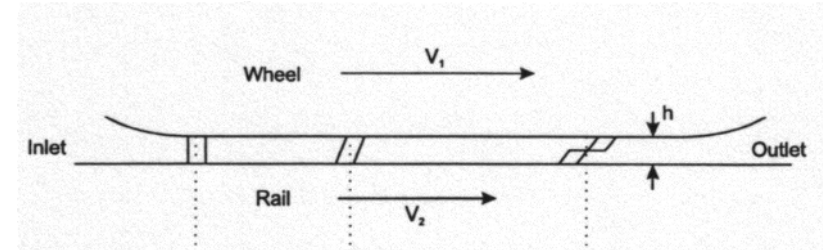


Getting “inside” the Wheel-Rail Interface



Source: <https://www.skodagroup.com/reference/metro-varsovia>

Getting “Inside” the Wheel Rail Interface



Kalousek, J, K Hou, E Magel, and K Chiddick. “The Benefits of Friction Management: A Third Body Approach.” In Proceedings of the World Congress on Railway Research (WCRR 1996), 8. Colorado Springs, 1996.



Fundamental #1: Contact Pressure & Geometry

The “contact patch”

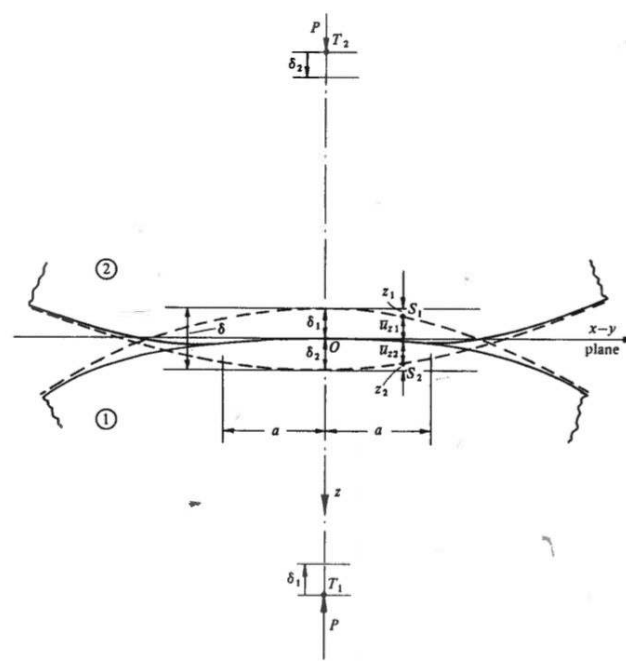
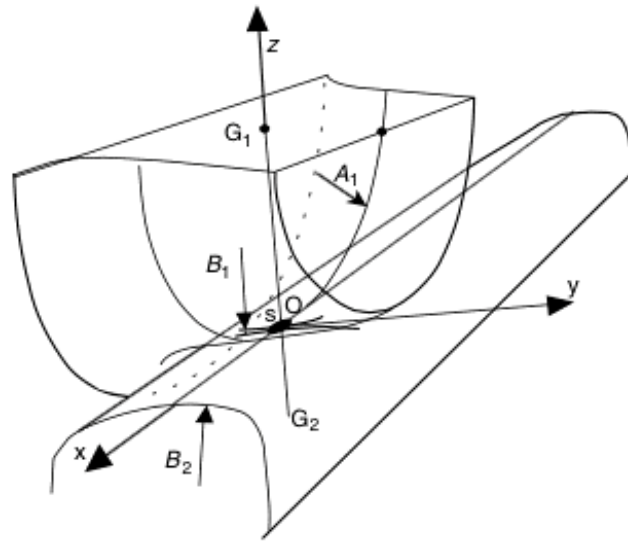
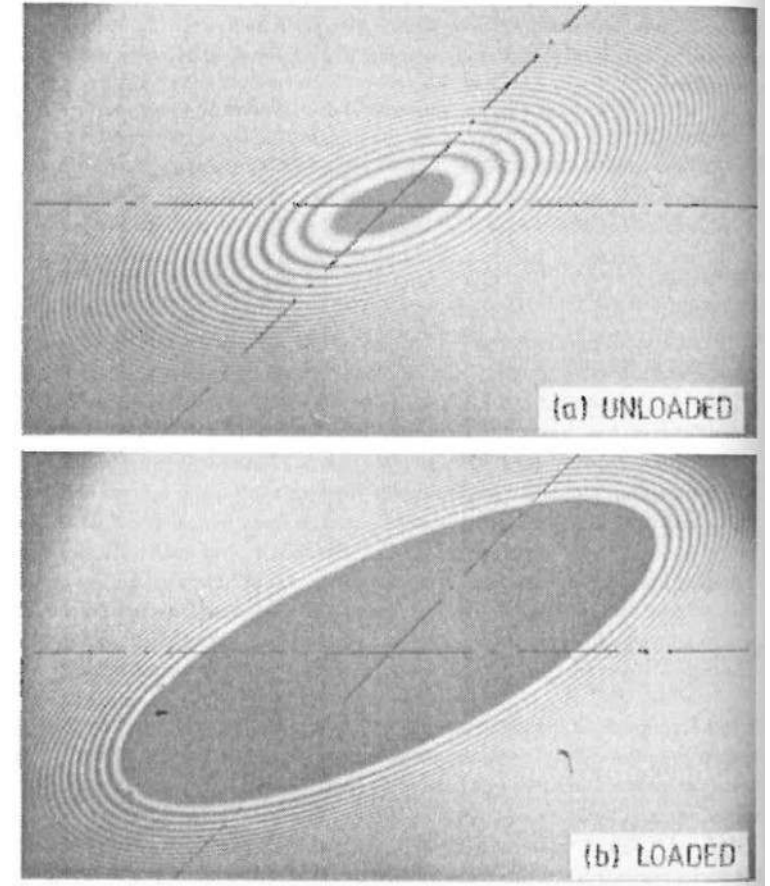
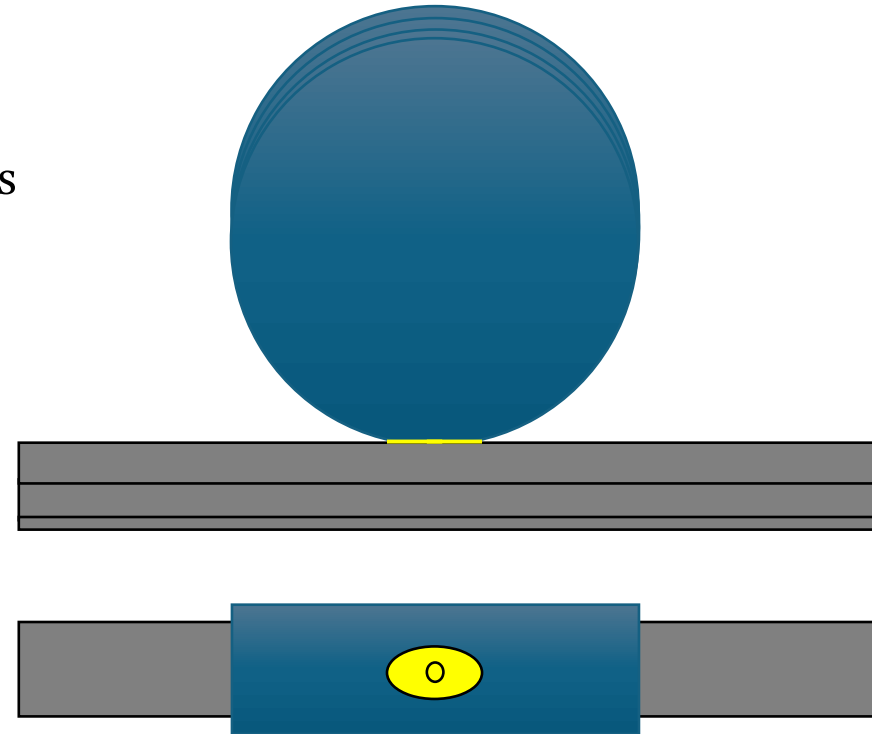
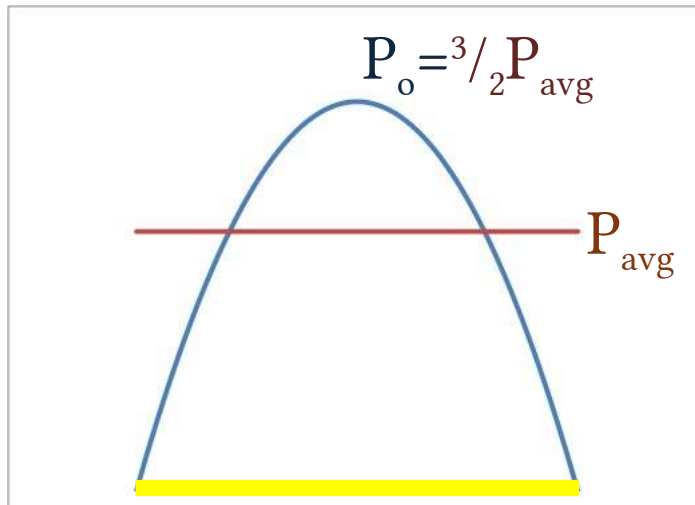


Fig. 4.1. Interference fringes at the contact of two equal cylindrical lenses with their axes inclined at 45° : (a) unloaded, (b) loaded.



Hertzian Contact

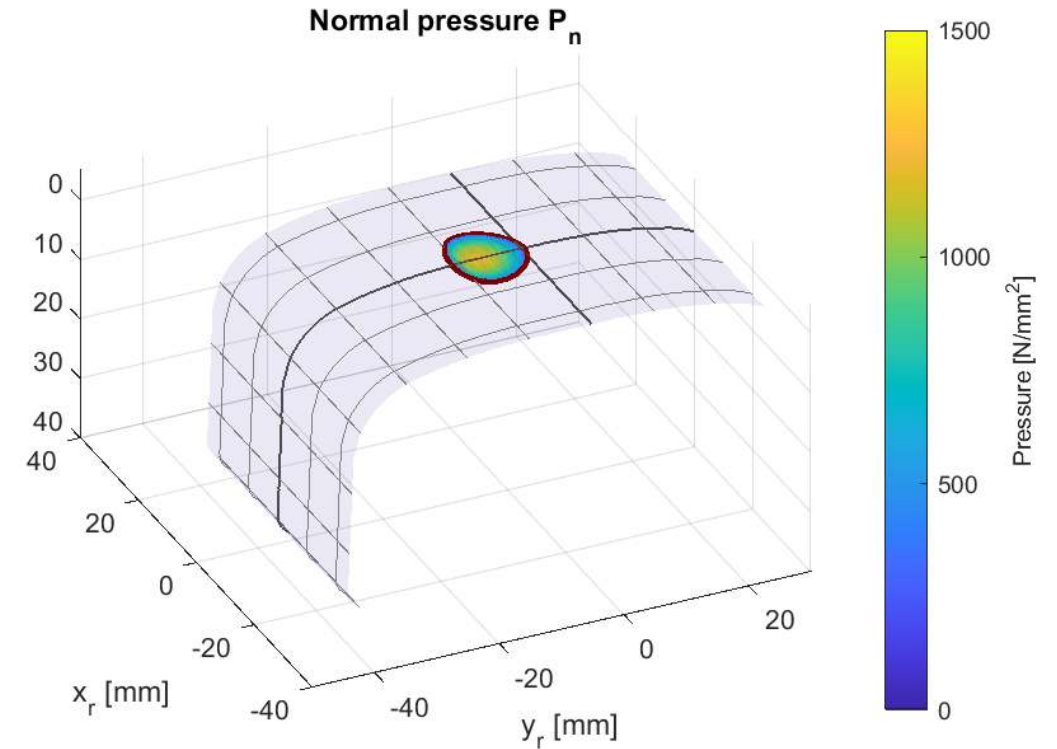
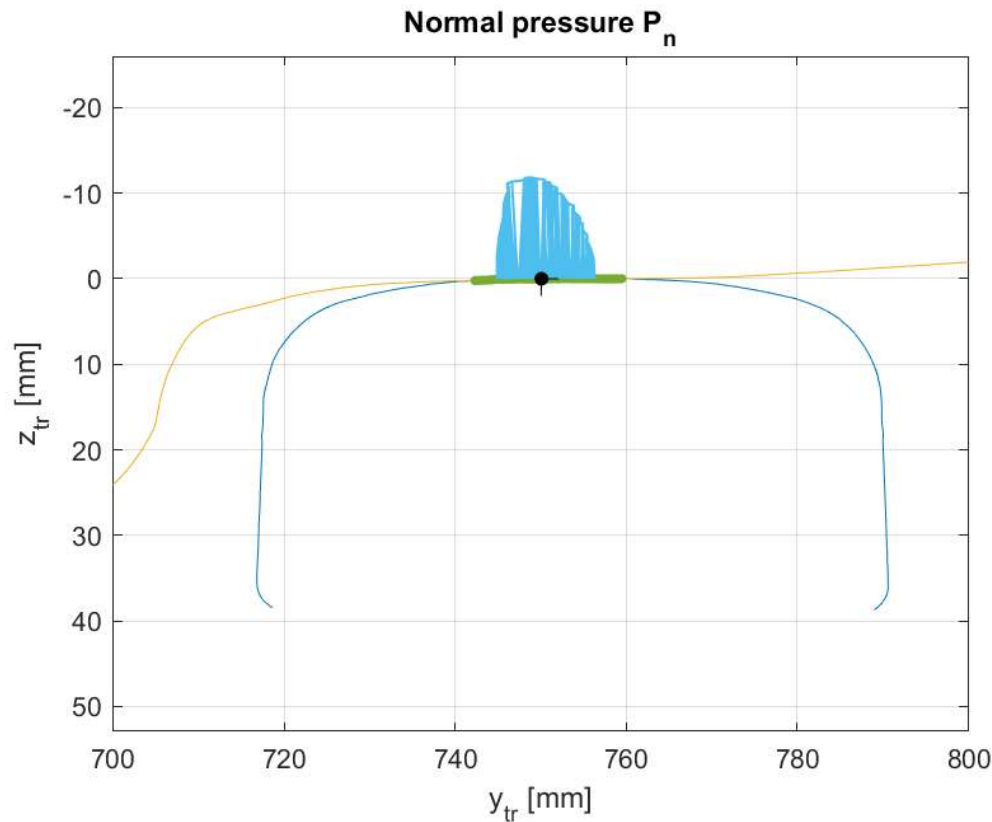
- Hertzian Contact (1881) describes the pressures, stresses and deformations that occur when curved elastic bodies are brought into contact.
- “Contact Patches” tend to be **elliptical**
- This yields **parabolic** contact pressures



- Contact theory was subsequently broadened to apply to rolling contact (Carter and Fromm) with non-elliptical contact and arbitrary creepage (Kalker)



(Quasi) Hertzian Contact in Practice

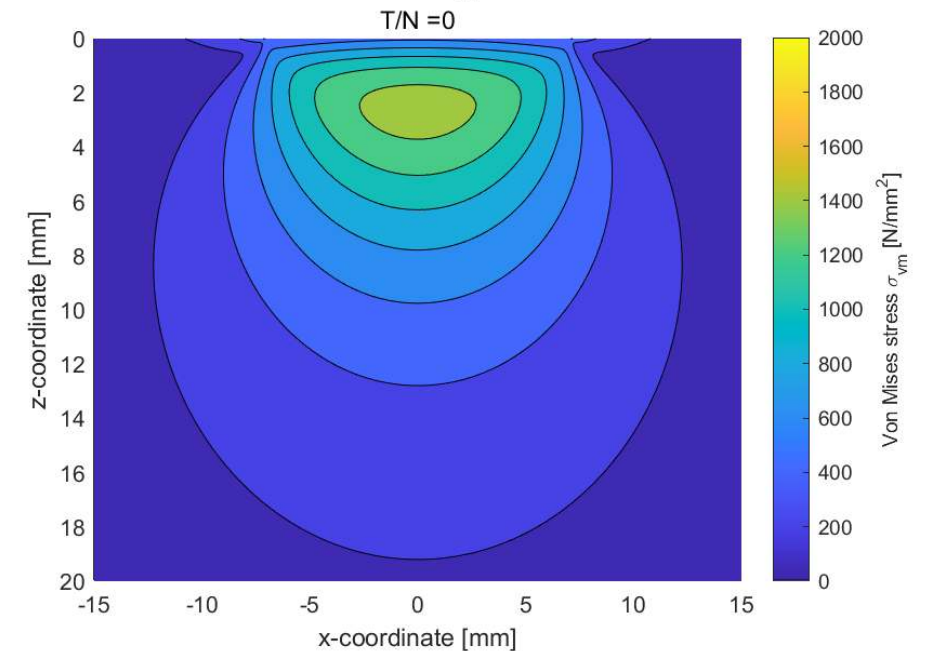
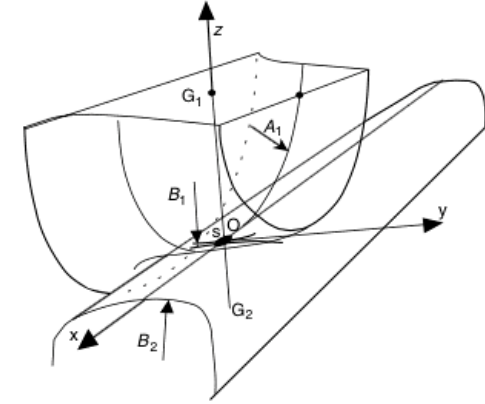


Vollebregt, E A H. "User Guide for CONTACT, Rolling and Sliding Contact with Friction." Technical Report, 2021.

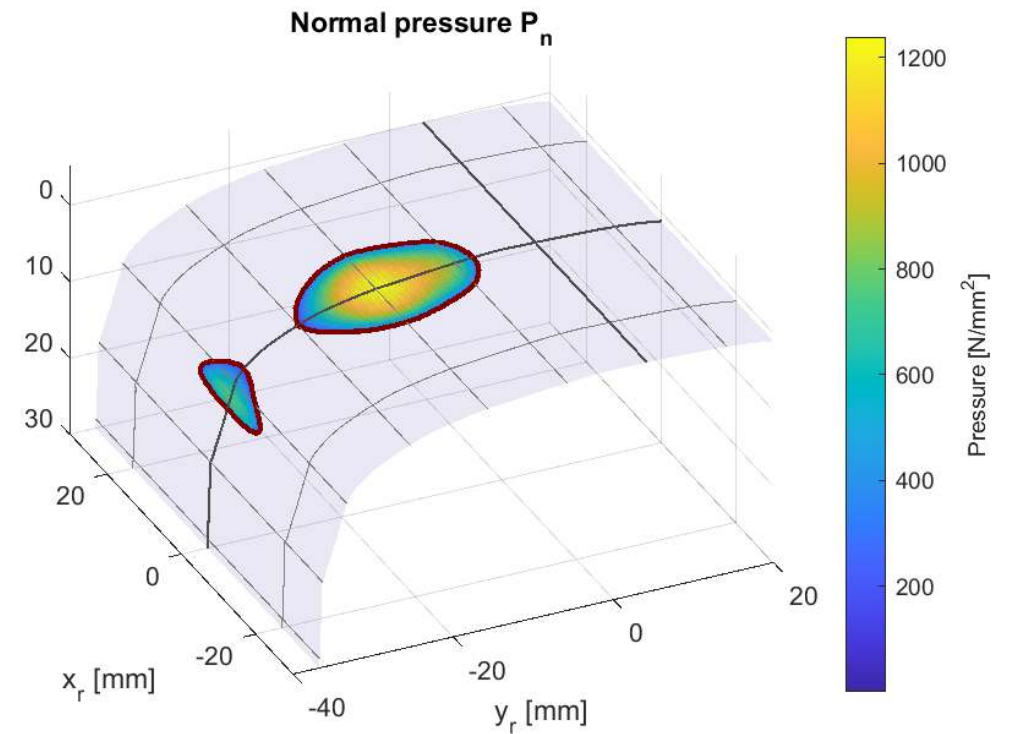
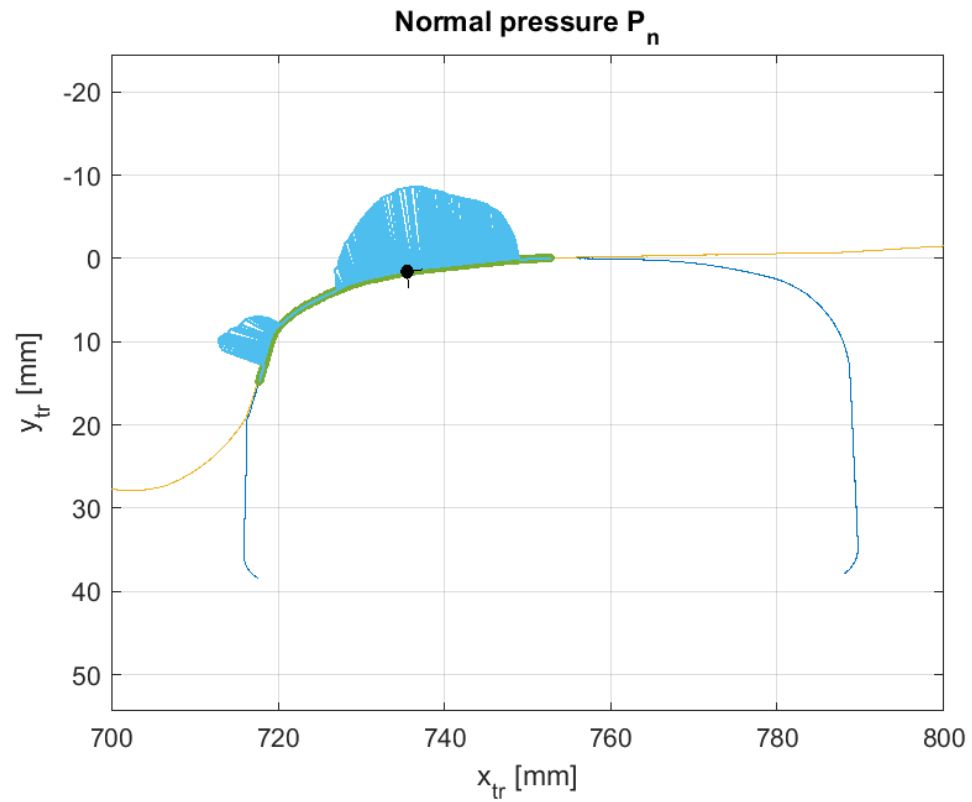


Hertzian Contact and Subsurface Stresses

- Example, using the *CONTACT Library* to calculate the characteristic (Von Mises) stresses for the following steel-on-steel contact conditions:
 - Radius of curvature of body 1: 0.5m (20”), aka the “wheel”
 - Radius of curvature of body 2: 0.25m (10”), aka the “rail”
 - Applied Force: 160kN (36kips), aka the “weight”



Non-Hertzian Contact



Vollebregt, E A H. "User Guide for CONTACT, Rolling and Sliding Contact with Friction." Technical Report, 2021.



Fundamental #2: Traction & Creepage



Creepage at the Wheel/Rail Interface

- Creepage within the contact patch can include three types:
 - Lateral
 - Longitudinal
 - Spin



What does Longitudinal Creepage *mean*?...



Coasting along a flat and dry highway

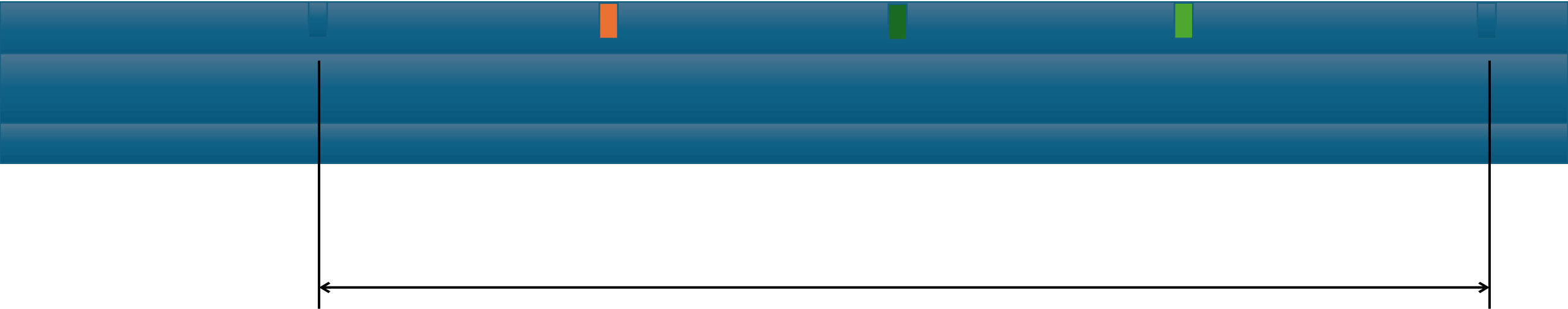
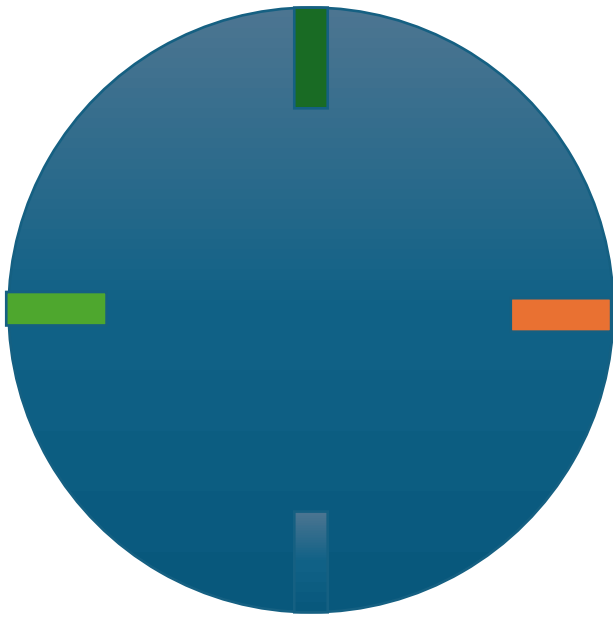


Climbing a steep and snowy mountain road



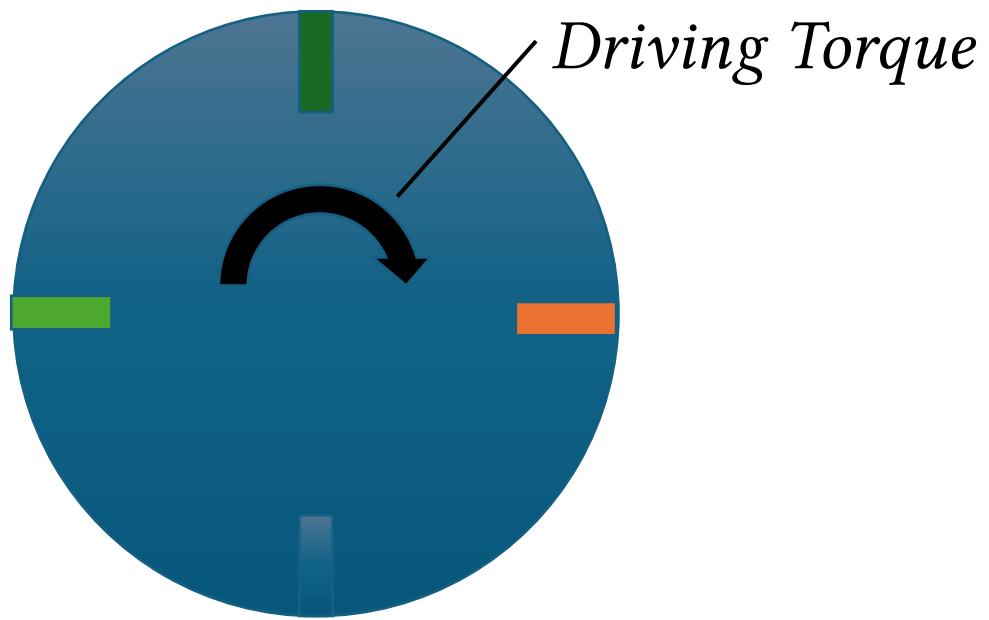
Free Rolling

*In free rolling, a wheel would rotate **100** times to travel a distance of **100** circumferences.*



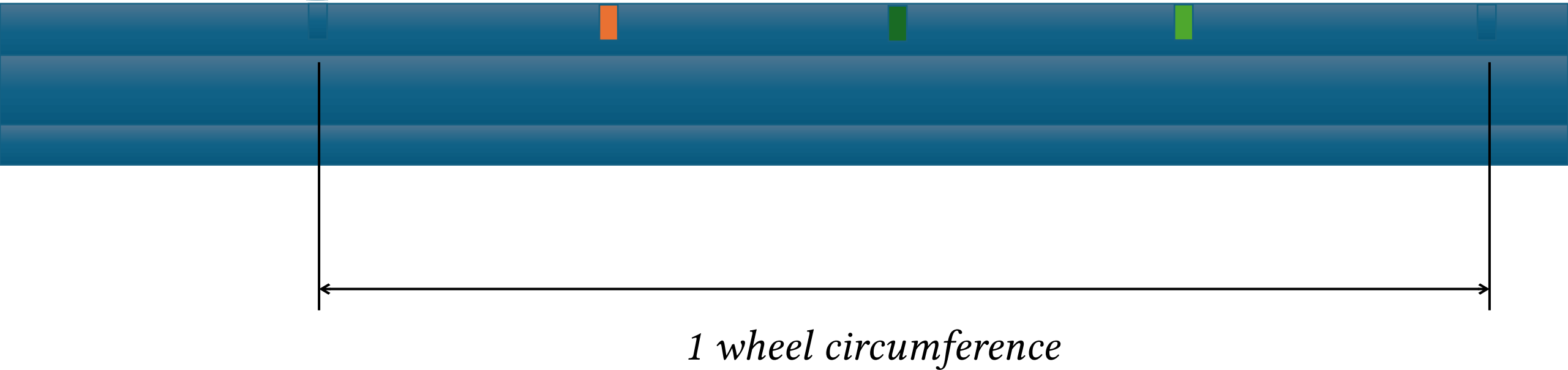
1 wheel circumference

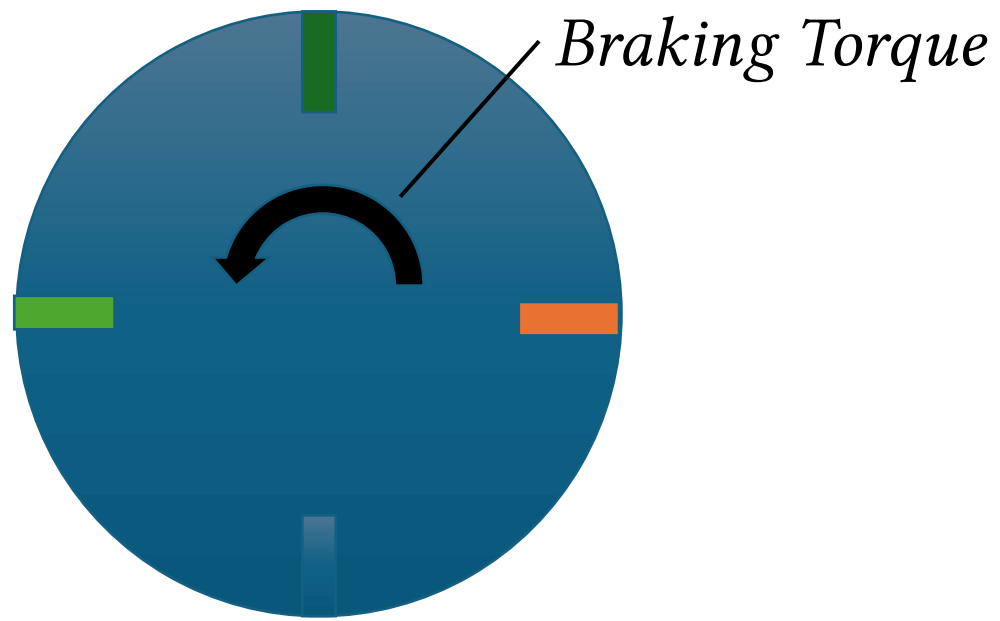




Positive (Longitudinal) Creepage

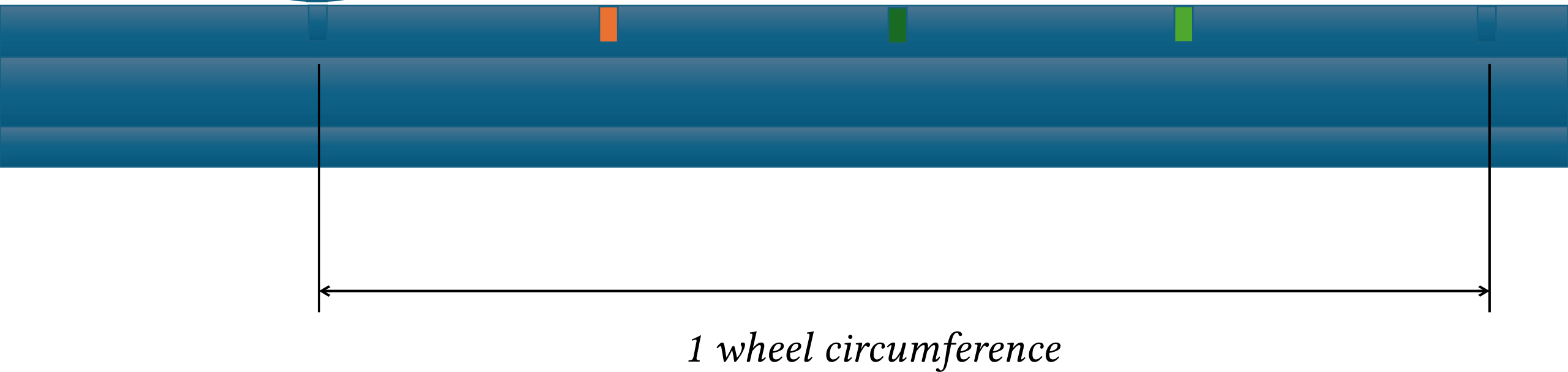
*At 1% **positive** creepage, a wheel would rotate **101** times to travel a distance of **100** circumferences.*



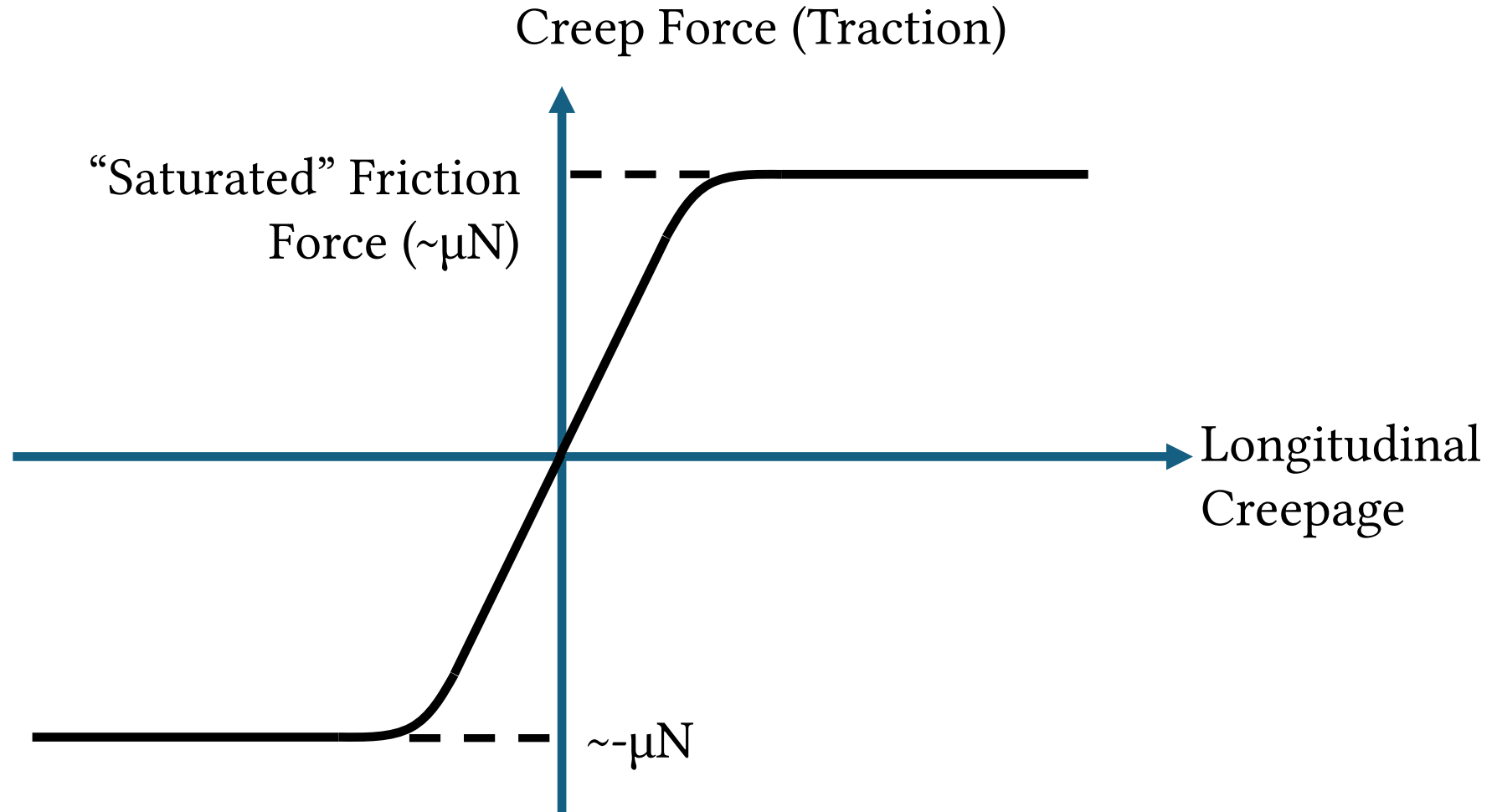


Negative (Longitudinal) Creepage

*At 1% **negative** creepage, a wheel would rotate **99** times to travel a distance of **100** circumferences.*

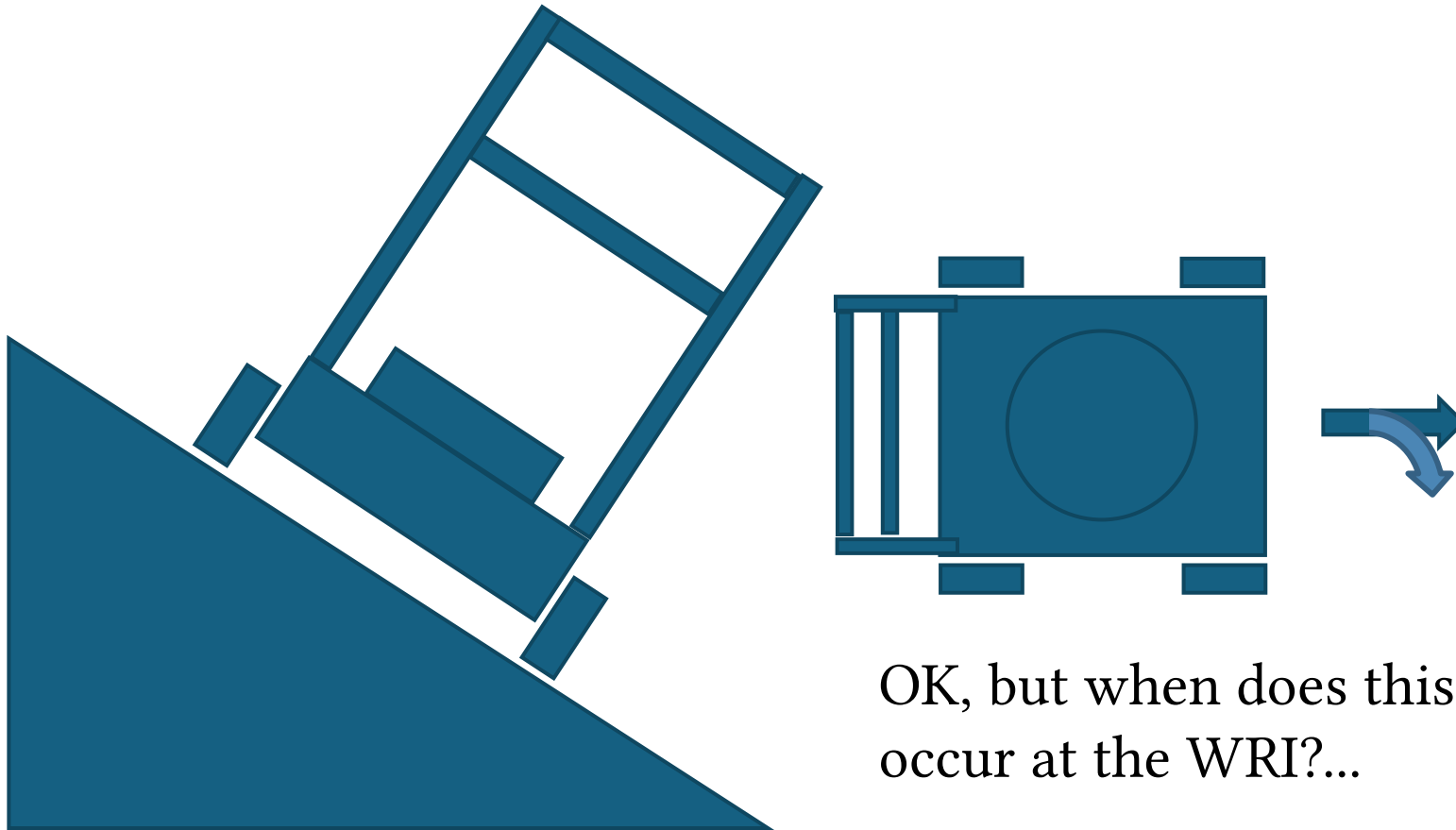


The Traction-Creepage Curve



Lateral creepage

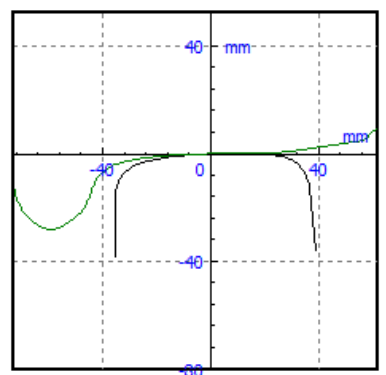
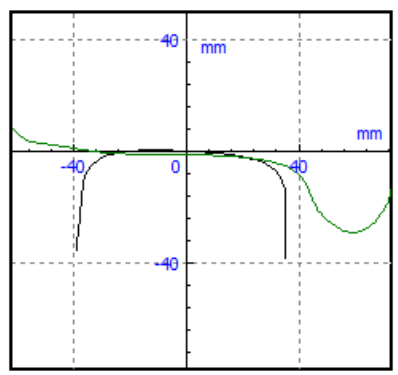
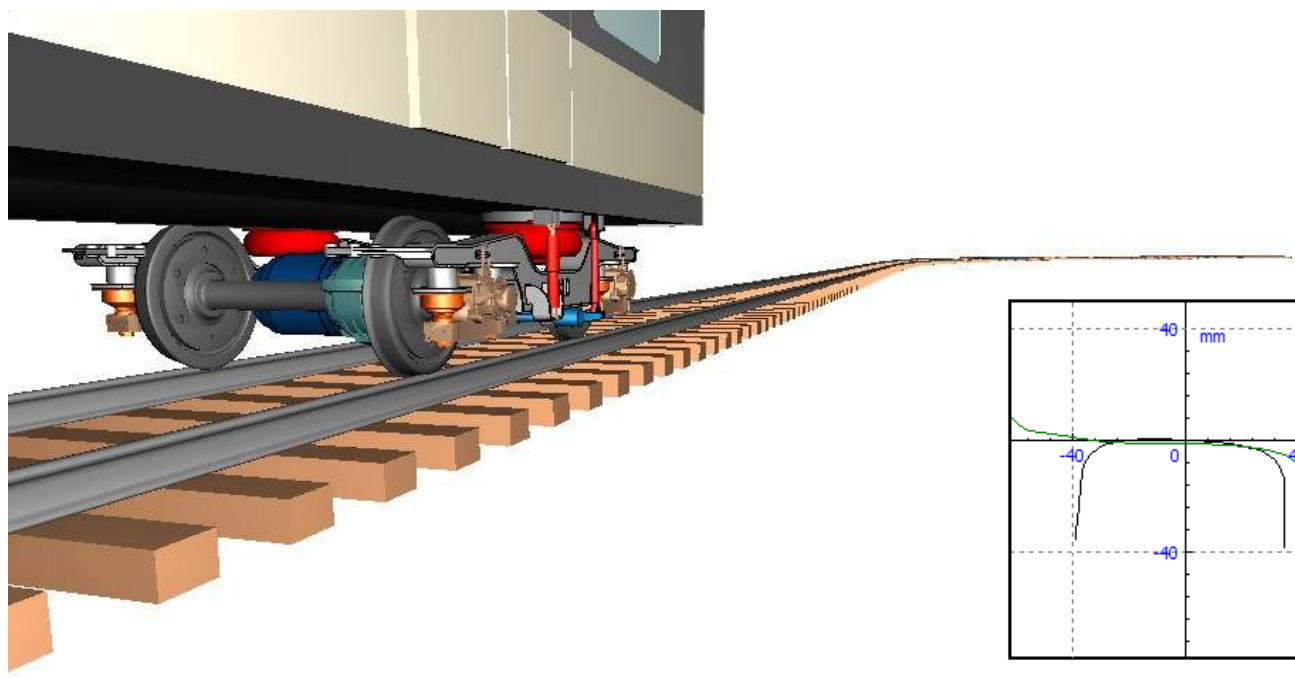
Imagine pushing a lawnmower across a steep slope...



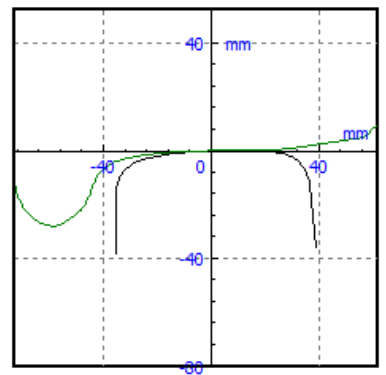
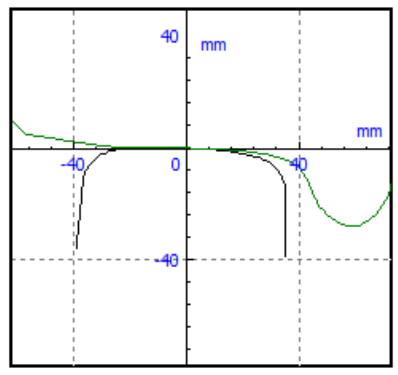
OK, but when does this occur at the WRI?...



Vehicle Curving (UM Simulation)



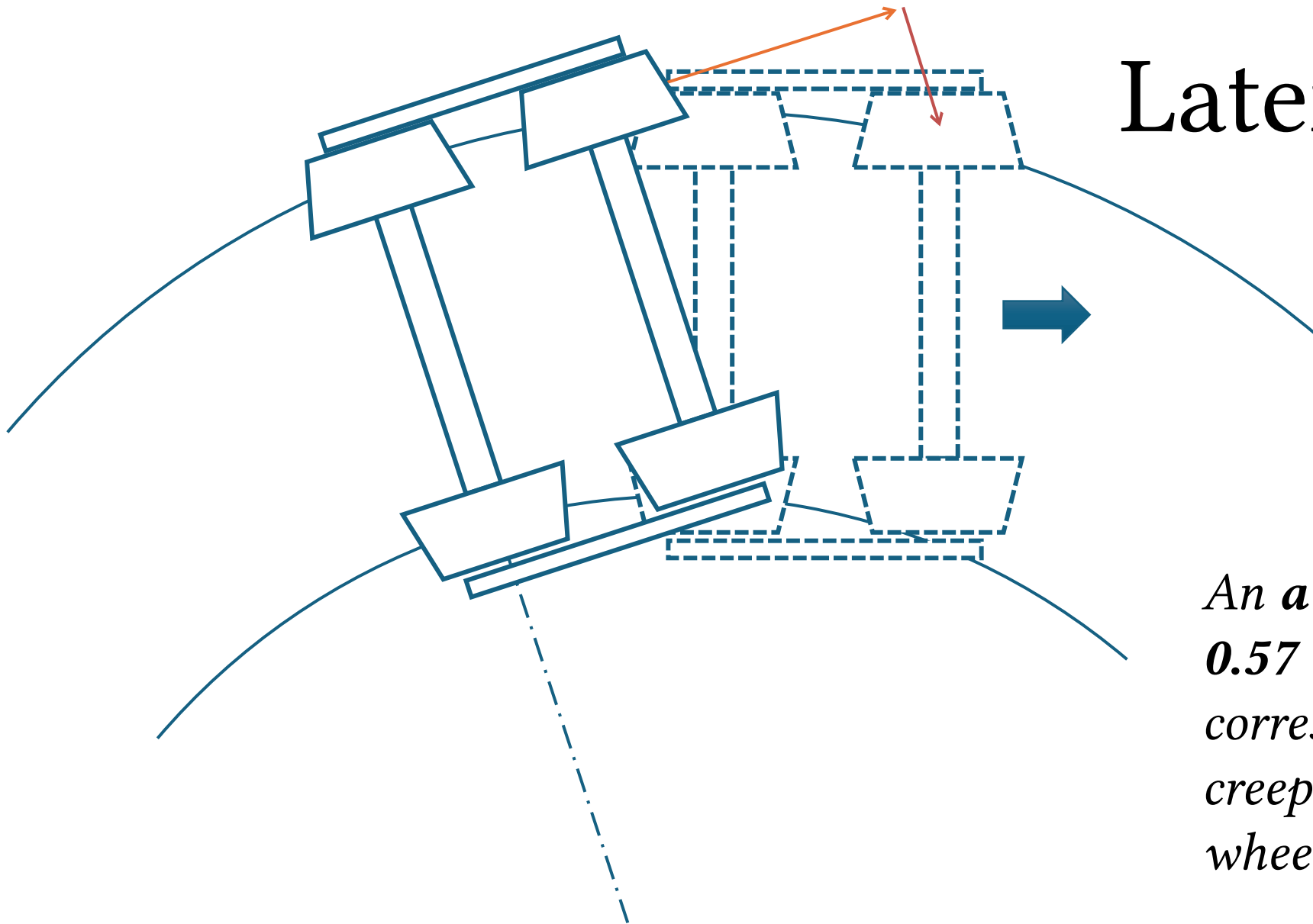
Leading Wheelset



Trailing Wheelset



Lateral Creepage

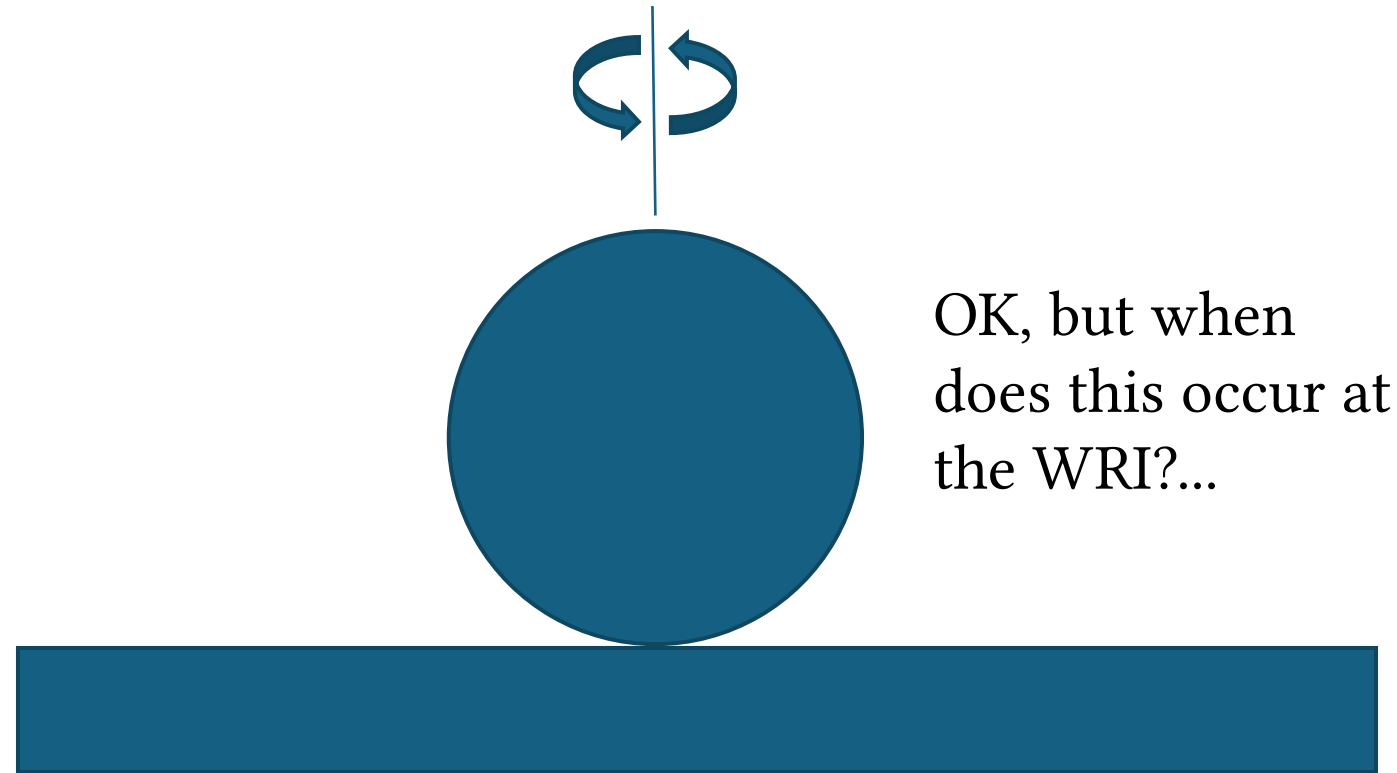


*An **angle of attack** (AoA) of **0.57 degrees** (0.01 Radians) corresponds to a lateral creepage of **1%** at the leading wheelset.*



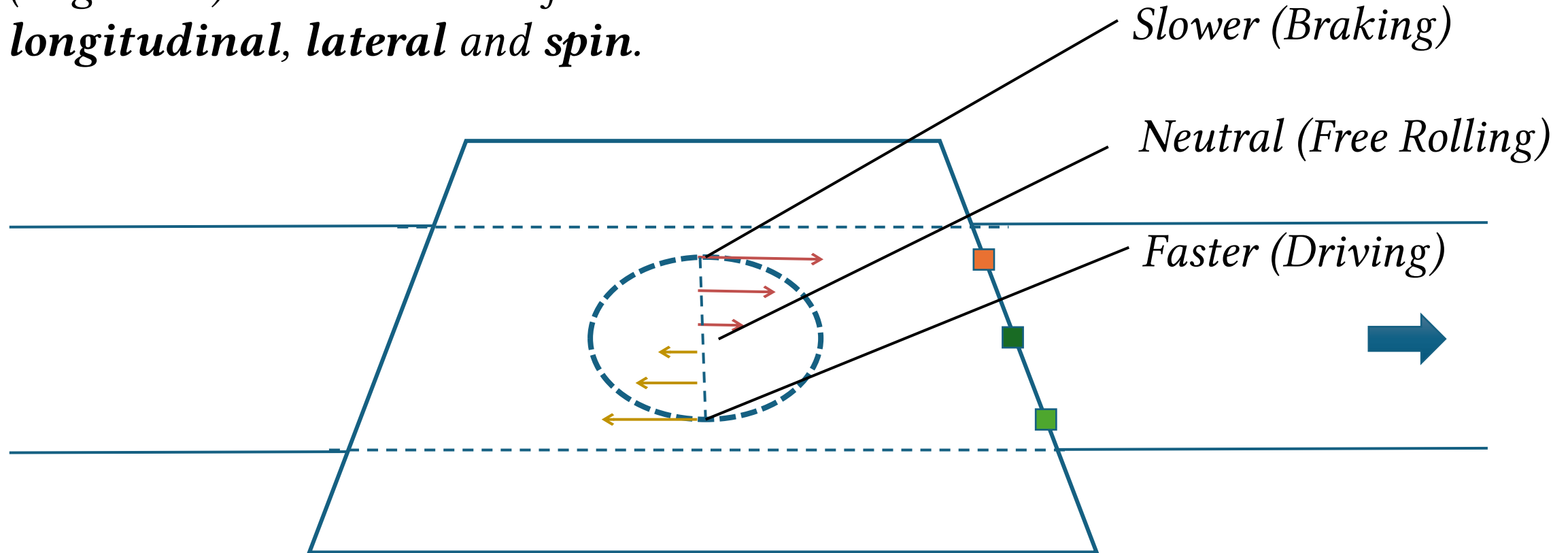
Spin Creepage

Think of spinning a coin on a tabletop....

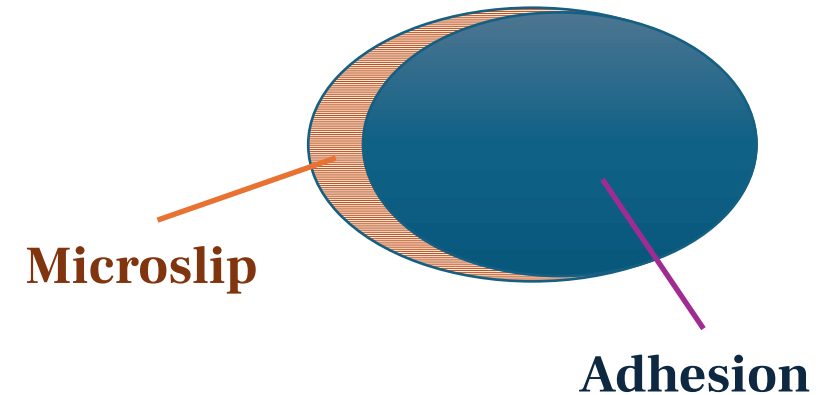
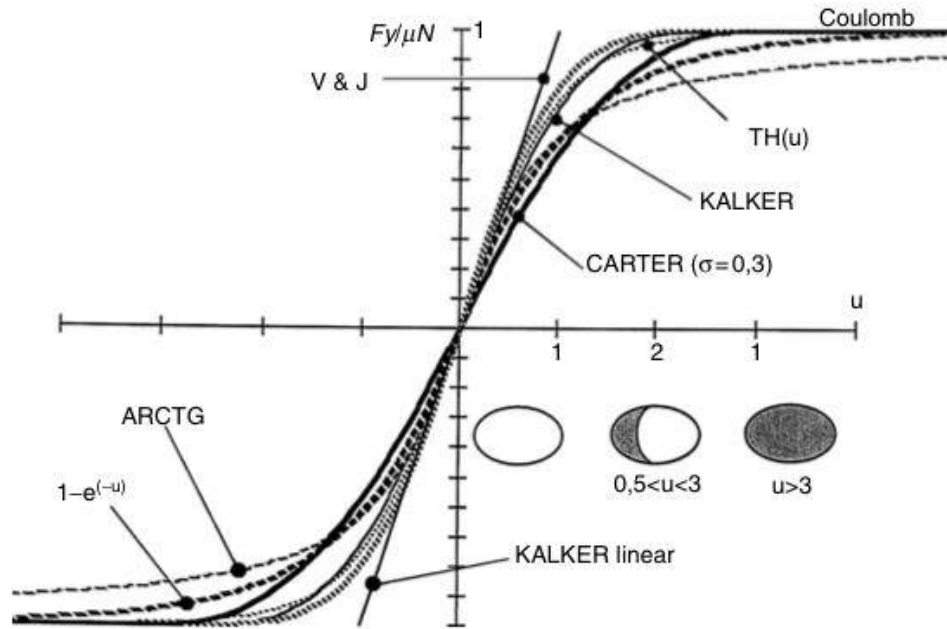


Spin Creepage

The **net creepage** vector at the wheel/rail interface is
(in general) a combination of
longitudinal, lateral and spin.



The Traction-Creepage Curve: Stick and Slip

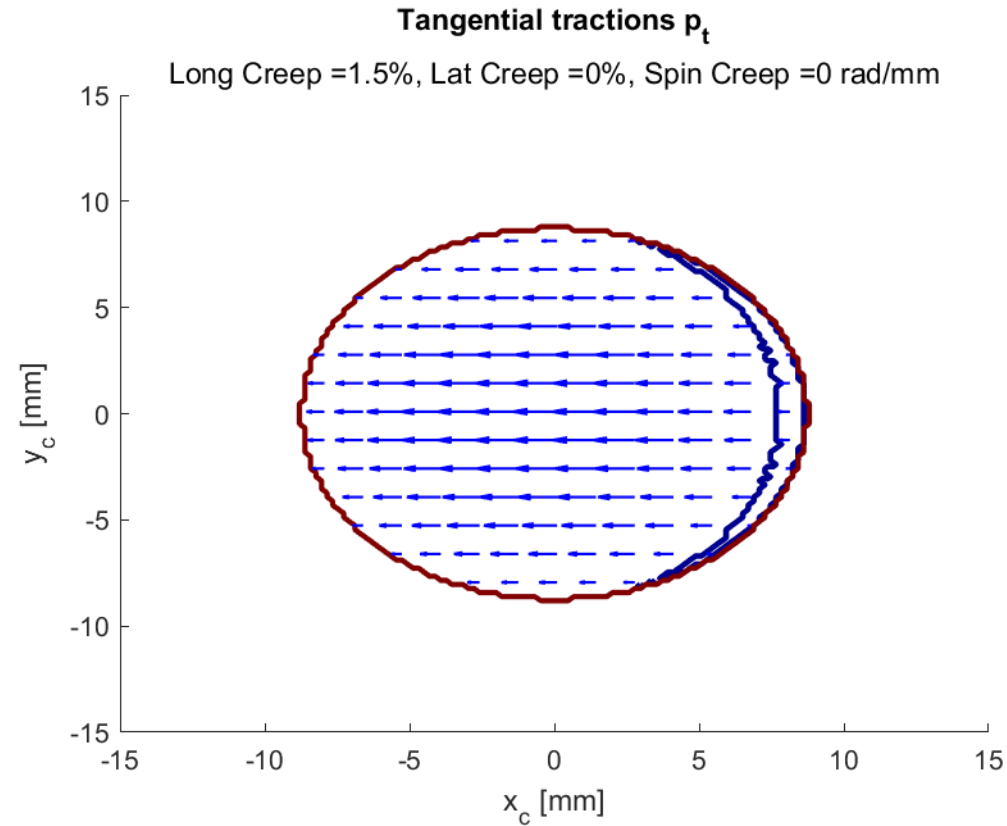


“Heuristic” expressions used for the saturation and physical meaning of the different parts.

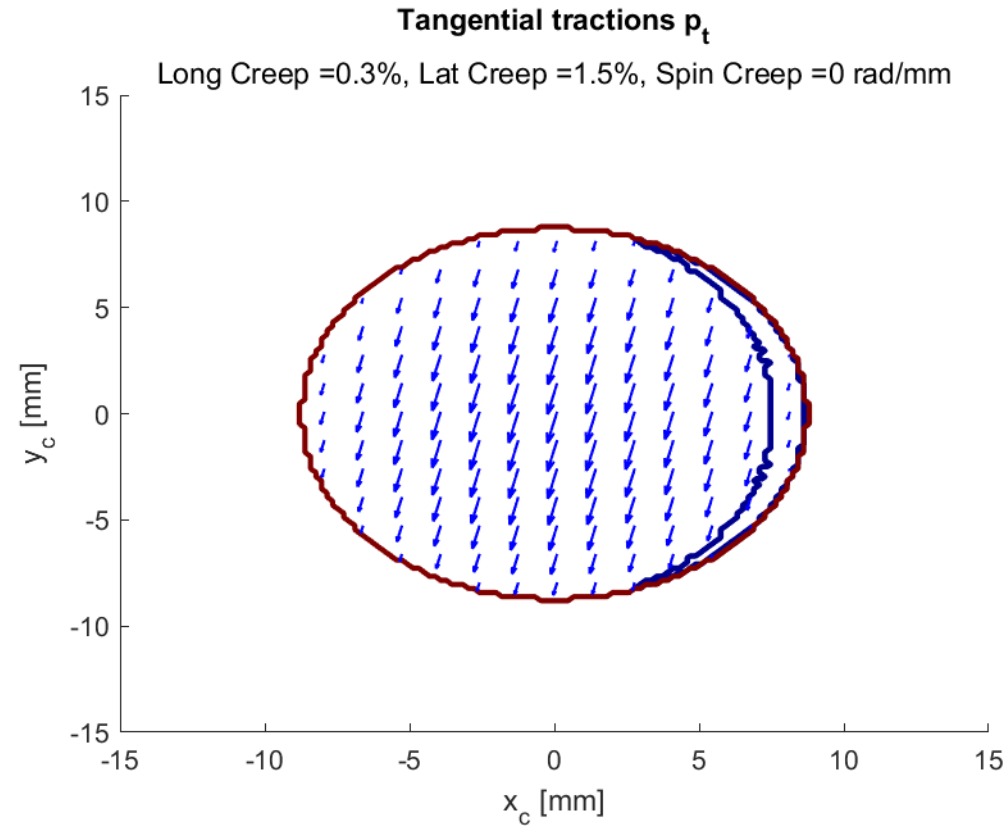
Iwnicki, S. (2006) Handbook of Railway Dynamics, CRC Press



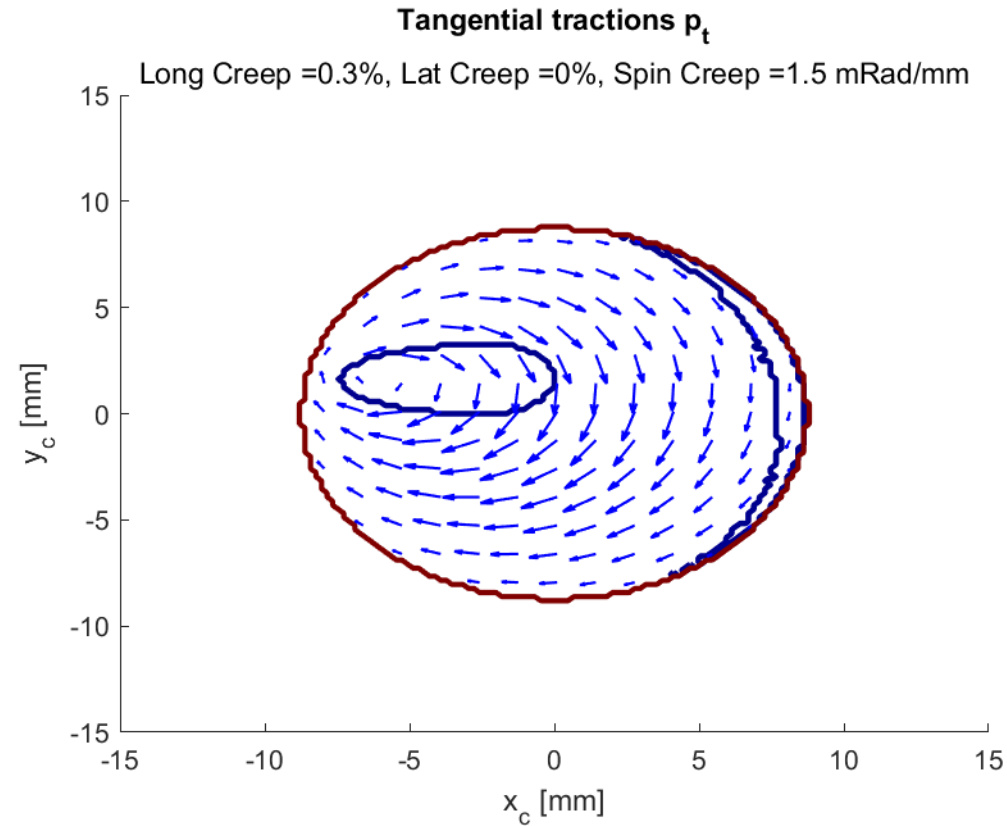
Increasing Longitudinal Creepage in the Contact Patch:



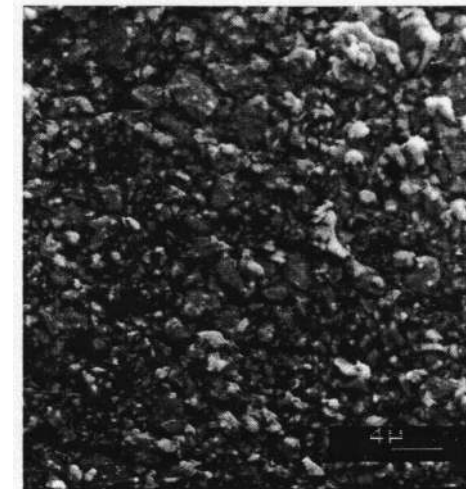
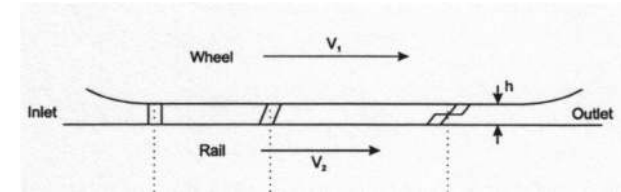
Combined Longitudinal and Lateral Creepage in the Contact Patch:



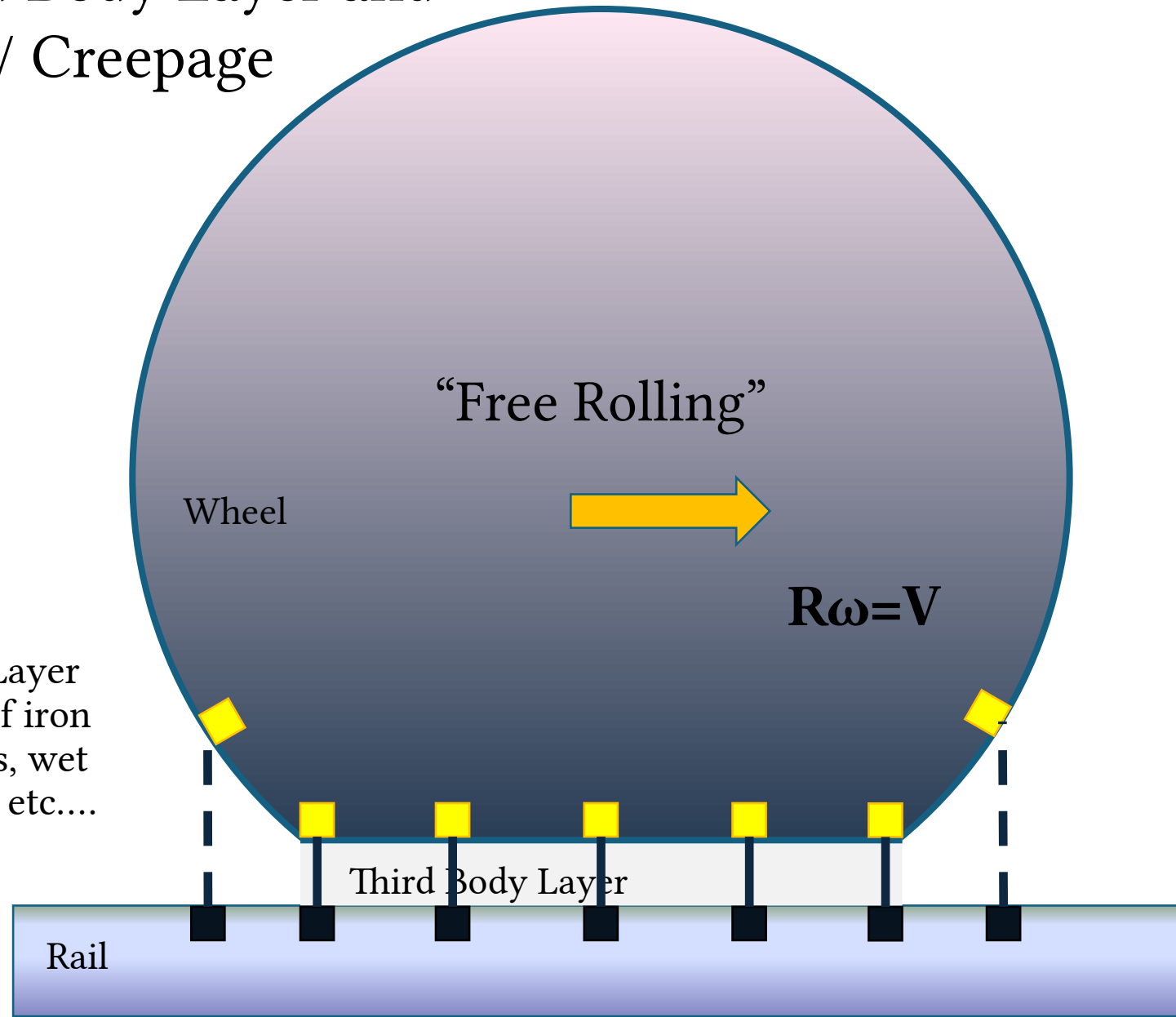
Combined Longitudinal and Spin Creepage in the Contact Patch:



Reality Check: The Third Body Layer

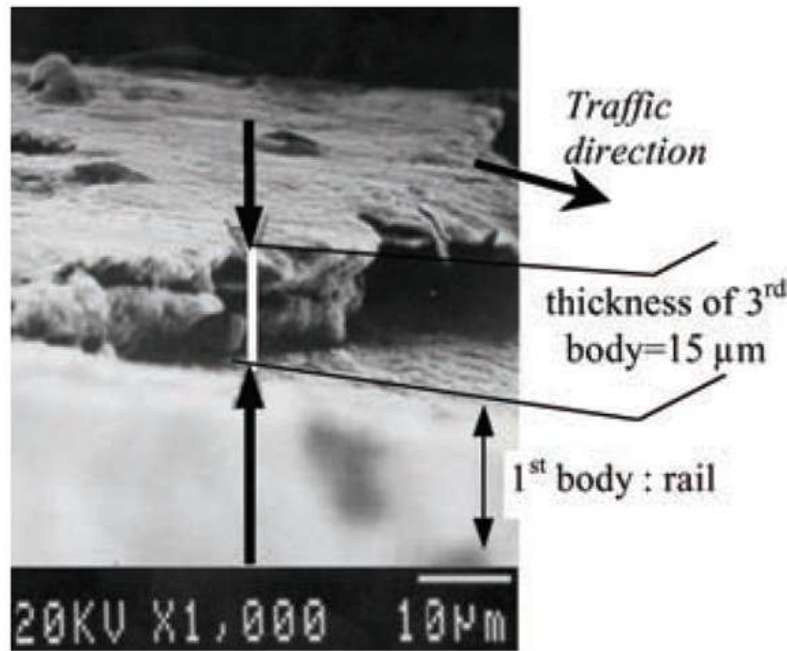


The Third Body Layer and Traction / Creepage

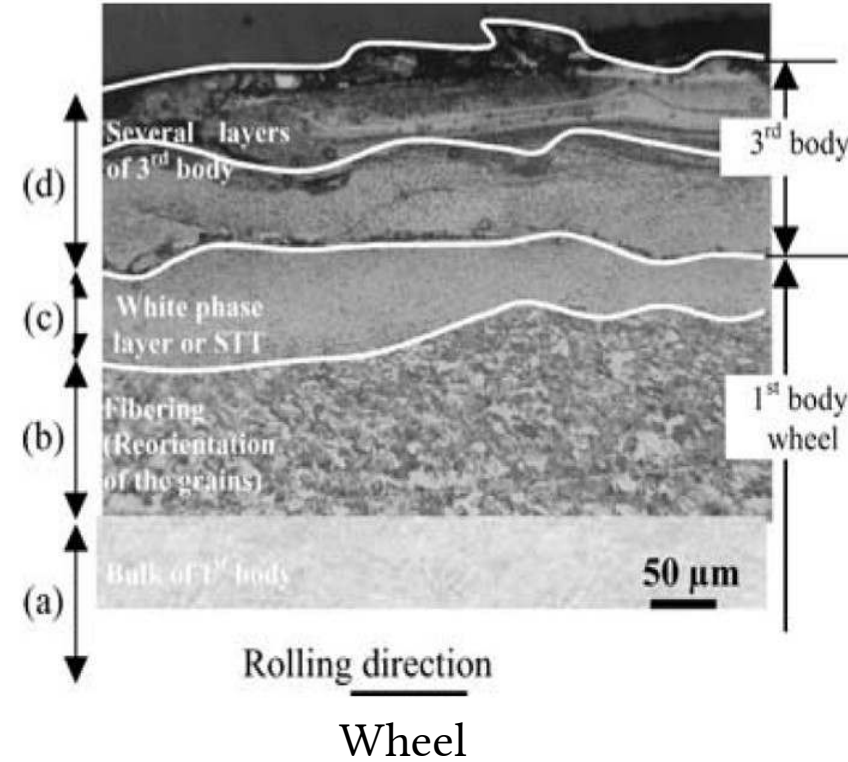


- Third Body Layer is made up of iron oxides, sands, wet paste, leaves etc....

Third Body Layer – Micron Scale



Rail
1

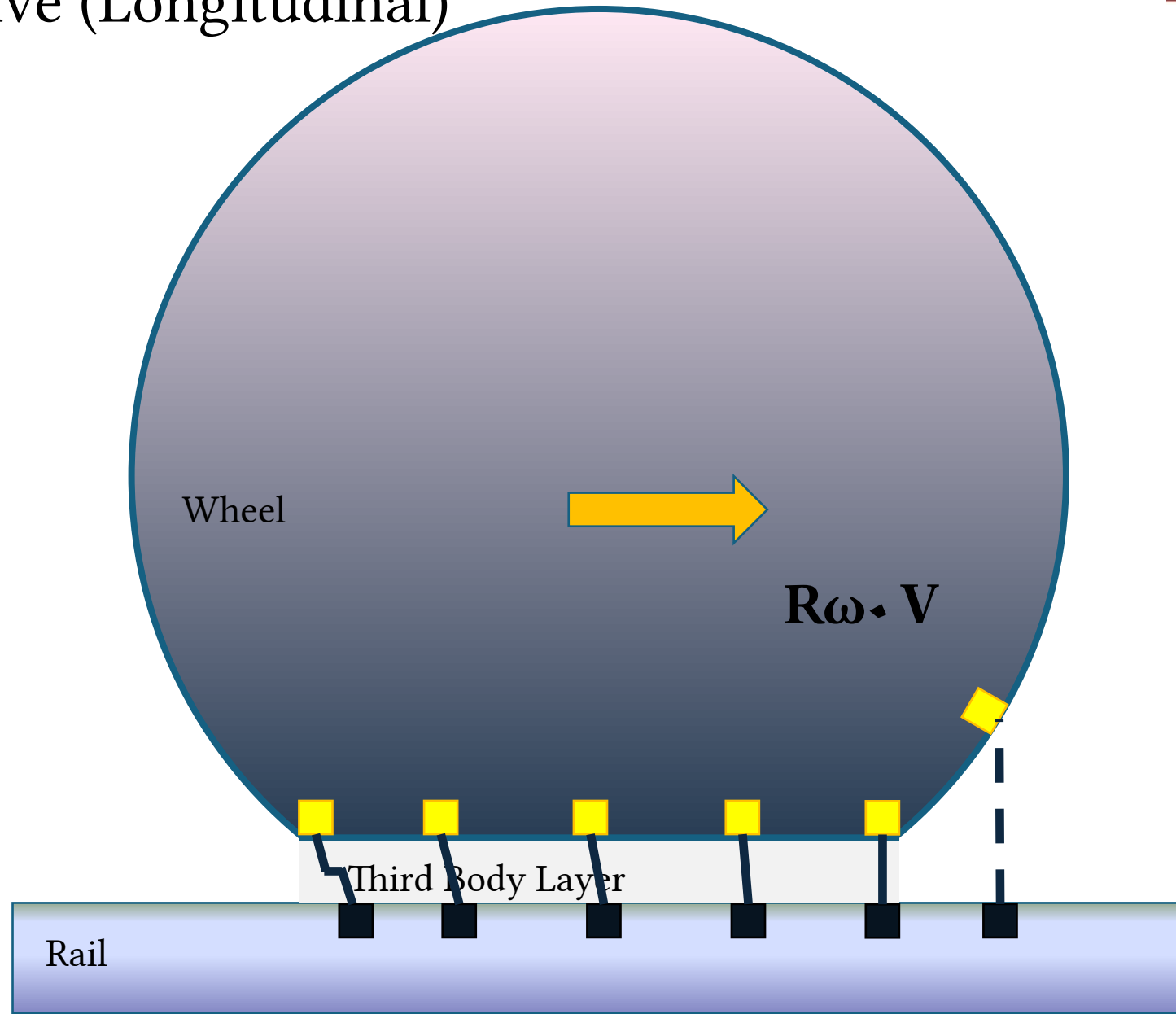


Wheel

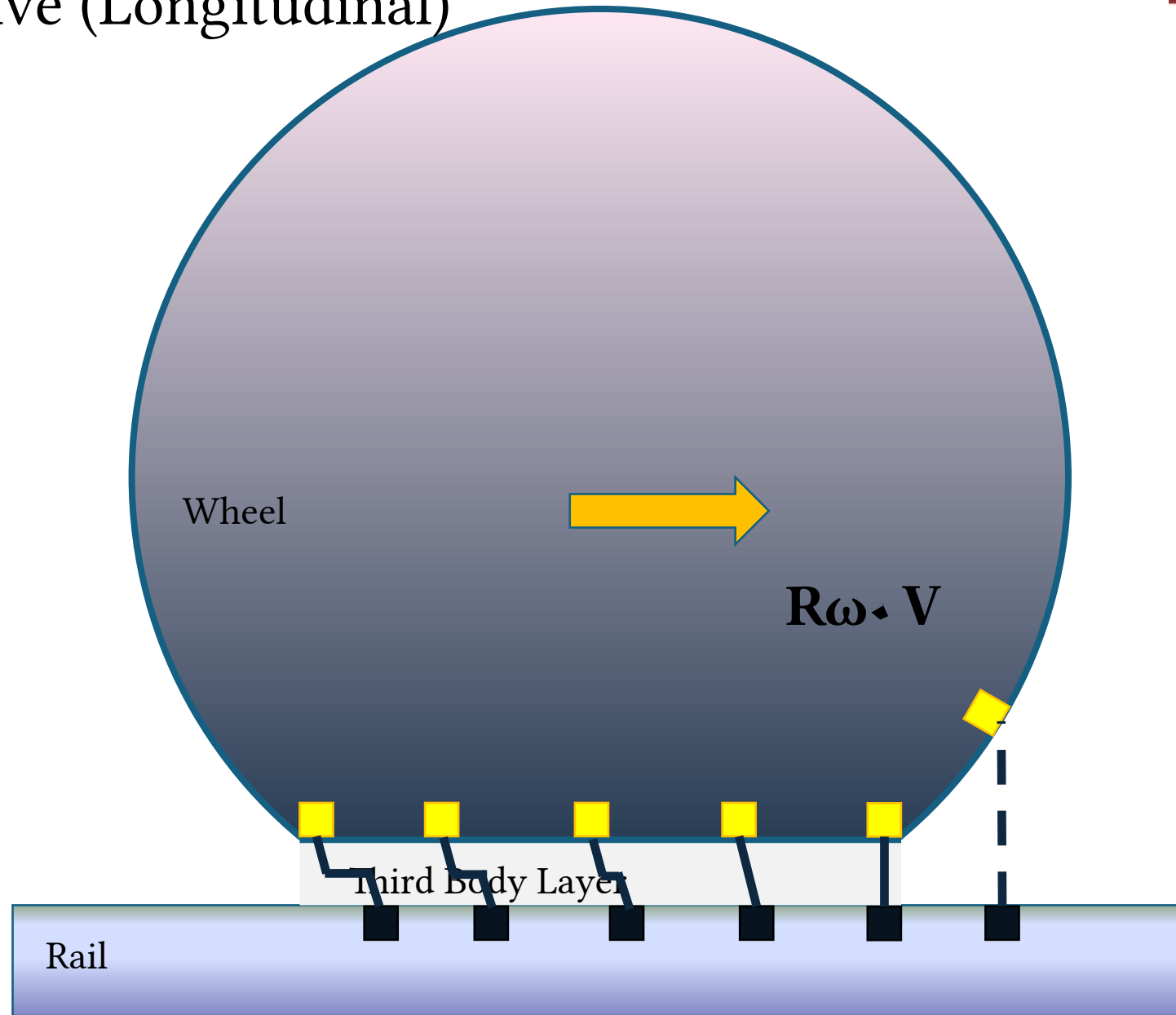
Y.Berthier, S. Decartes, M.Busquet et al. (2004). The Role and Effects of the third body in the wheel rail interaction. *Fatigue Fract. Eng. Mater Struct.* 27, 423-436



“Small” Positive (Longitudinal) Creepage



“Large” Positive (Longitudinal) Creepage

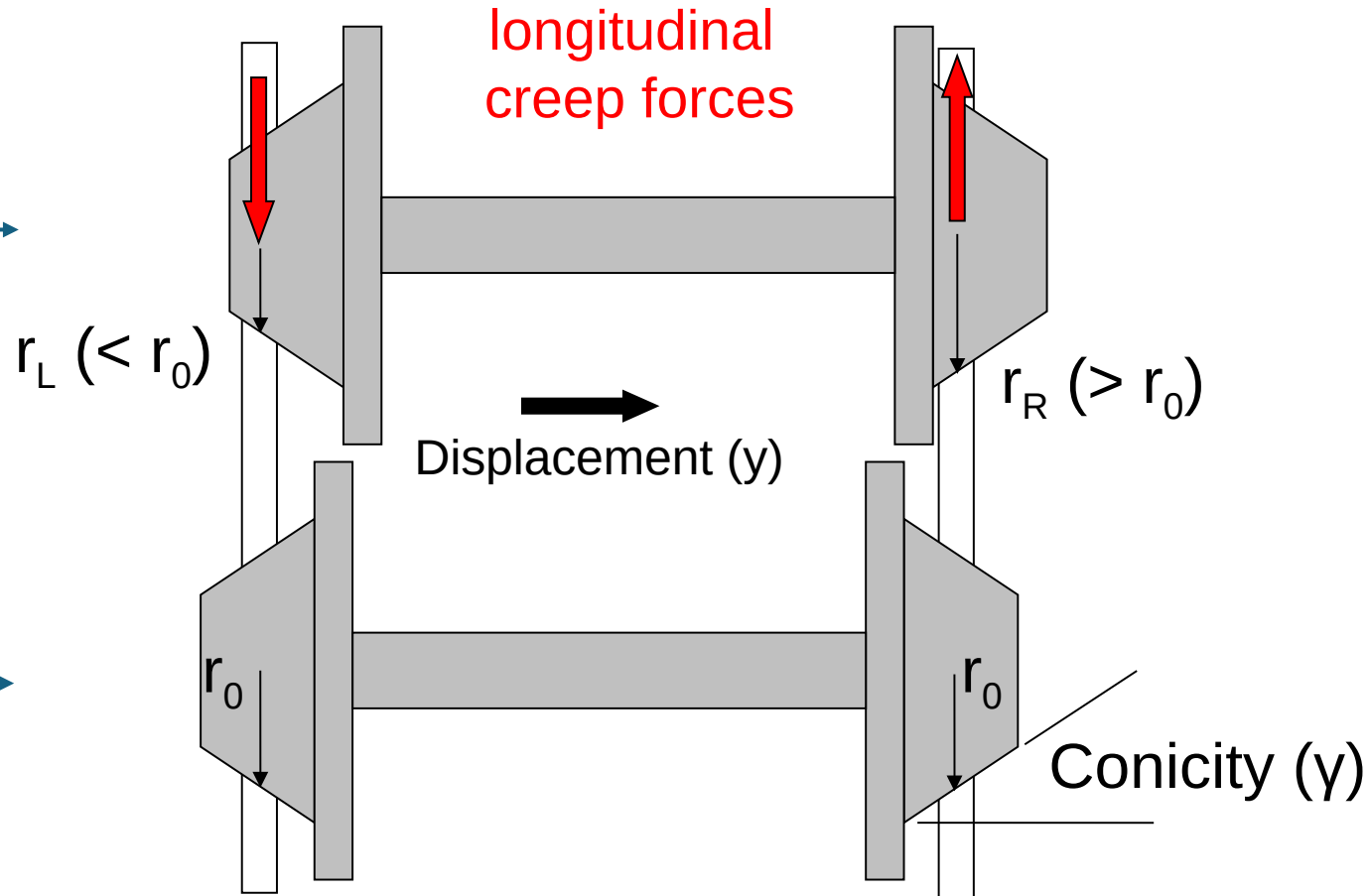
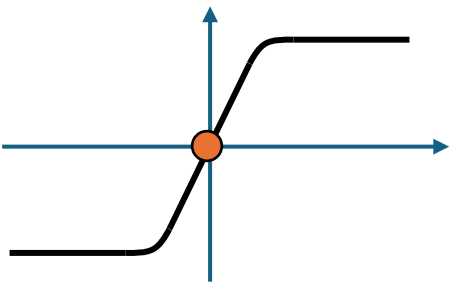
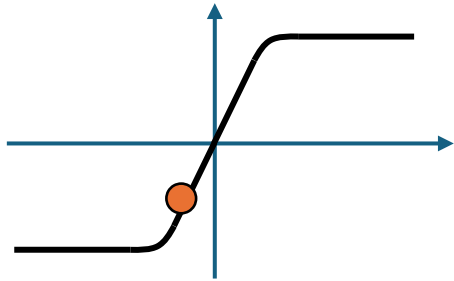


Steering Forces: Putting it all together

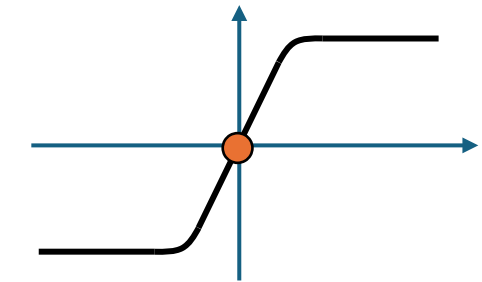
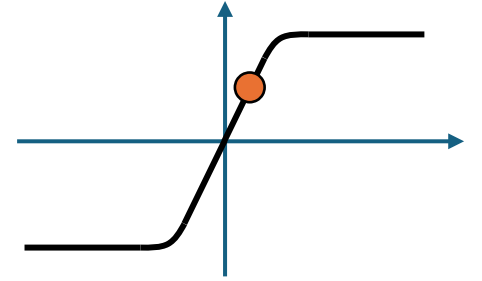


The Wheelset and Steering Forces

longitudinal
traction/creepage

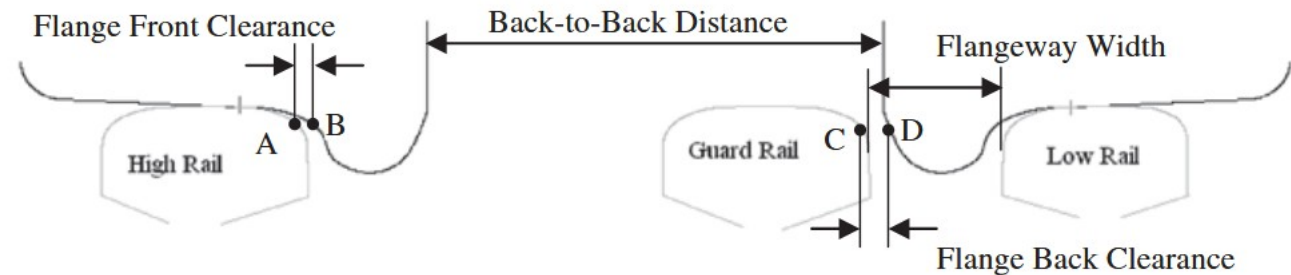


longitudinal
traction/creepage



A Special Case: Wheel-Rail contact and Restraining (Guard) Rails

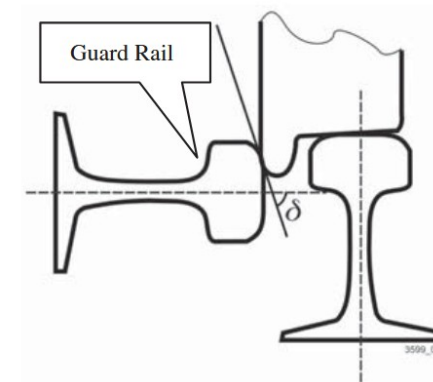
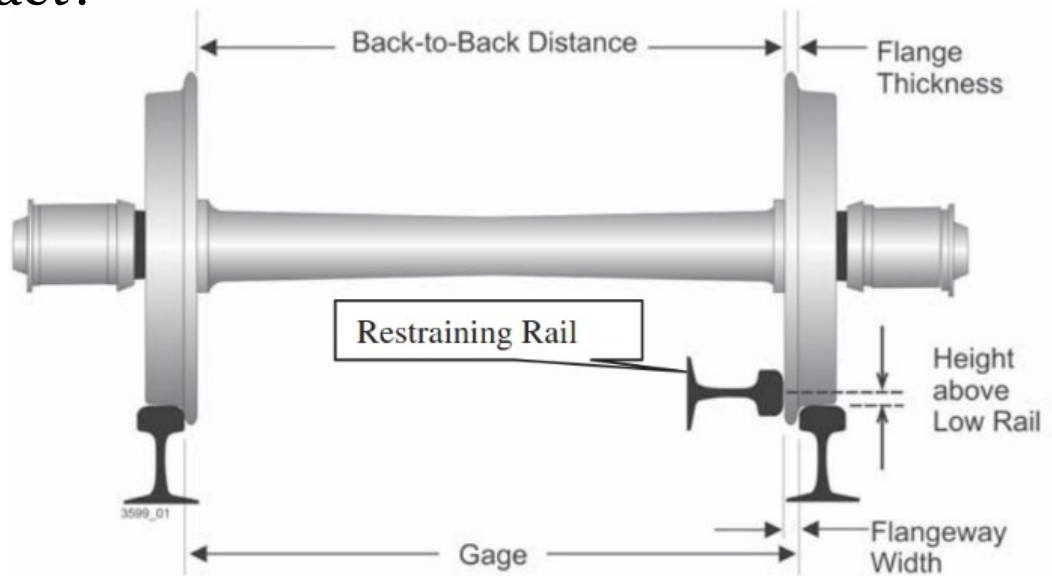
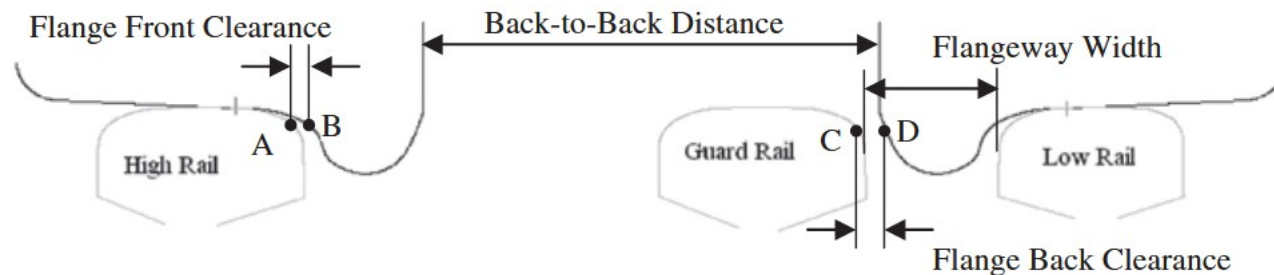
- Restraining (Guard) Rails are used in transit systems to:
 - Reduce rail wear (esp. high rail wear) in curves
 - Increase the track's resistance to wheel-climb derailment.



Transit Cooperative Research Program, Transportation Research Board, and National Academies of Sciences, Engineering, and Medicine. Use of Guard/Girder/Restraining Rail. Transportation Research Board, 2007.

Wheel-Rail contact and Restraining (Guard) Rail Geometry

- How would choices in restraining rail geometry (e.g. flange clearance and guard rail height) affect wheel rail contact?



Transit Cooperative Research Program, Transportation Research Board, and National Academies of Sciences, Engineering, and Medicine. Use of Guard/Girder/Restraining Rail. Transportation Research Board, 2007.



Conclusions

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- Reality Check: The Third Body Layer
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Thank You

